

PES UNIVERSITY

Department of Computer Science & Engineering



Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #05 — Academic Elective Bidding &
Allocation System

Course: Software Engineering Lab

DELIVERABLE 1

Requirements Table

System: Academic Elective Bidding & Allocation System

Actors: Student, Academic Registrar, Instructor

Context: University departments need an automated elective course selection mechanism where students allocate bidding credits across preferences and the engine solves allocation constraints while preventing timetable collisions.

Functional Requirements (FR-001 to FR-005)

REQ ID	TYPE	DESCRIPTION	PRIORITY	ACCEPTANCE CRITERIA	RATIONALE
FR-001	Functional	The system shall allow students to distribute 100 bidding credits across their ranked elective preferences and validate prerequisite course completions.	High	Pass: Credits deduct accurately and prerequisites validated. Fail: Total allocated credits exceed 100 or unmet prerequisites bypassed.	Core bidding functionality; ensures fair, merit-based allocation and enforces academic eligibility.
FR-002	Functional	The system shall display a catalogue of available elective courses showing course name, instructor, schedule, available seats, and prerequisite requirements.	High	Pass: Catalogue loads within 3 seconds and shows accurate, real-time seat counts matching the database. Fail: Stale or missing course data displayed, or page fails to render.	Students need complete, up-to-date information to make informed bidding decisions.
FR-003	Functional	The system shall detect and prevent timetable collisions when a student bids on electives with overlapping class schedules, and display a clear warning identifying the conflicting courses.	High	Pass: Submitting bids for two overlapping-schedule courses triggers a collision warning and blocks submission. Fail: Overlapping bids are accepted without any alert.	Prevents students from being allocated to courses they cannot physically attend.
FR-004	Functional	The system shall execute an allocation engine that processes all submitted student bids, optimises seat assignments based on credit weightings, and publishes final allocation results upon completion.	High	Pass: After the engine runs, every student receives allocations that respect seat capacity and credit rankings; results are visible in the portal. Fail: Over-allocation of seats or unprocessed bids remain.	Central engine that converts bids into fair, constraint-satisfying course assignments.

REQ ID	TYPE	DESCRIPTION	PRIORITY	ACCEPTANCE CRITERIA	RATIONALE
FR-005	Functional	The system shall allow the Academic Registrar to configure bidding parameters including bidding window open/close dates, per-course seat capacities, and maximum credits allocable to a single course.	Medium	Pass: Registrar updates bidding window dates and seat cap; changes reflect immediately in the student-facing interface. Fail: Configuration changes are not persisted or not visible to students.	Gives the registrar administrative control to adapt the bidding cycle to institutional needs.

Non-Functional Requirements (NFR-001 to NFR-002)

REQ ID	TYPE	DESCRIPTION	PRIORITY	ACCEPTANCE CRITERIA	RATIONALE
NFR-001	Performance	The final elective allocation solver shall process 5,000 student bids and resolve schedule conflicts in under 30 seconds.	High	Pass: Benchmarking tests confirm target latency under simulated peak load of 5,000 concurrent bids. Fail: Solver exceeds 30-second threshold or fails under concurrent load.	Ensures timely results so registration deadlines are met even at scale.
NFR-002	Security	The system shall enforce role-based access control so that students can only view and modify their own bids, and registrars have administrative privileges; all data in transit shall be encrypted using TLS 1.2 or higher.	High	Pass: Penetration test confirms no cross-user data leakage; TLS certificate validation passes; unauthorised API calls return HTTP 403. Fail: A student can view/modify another student's bids, or data is transmitted in plaintext.	Protects sensitive academic and personal data and complies with university data-governance policies.

DELIVERABLE 2

UML Use-Case Diagram

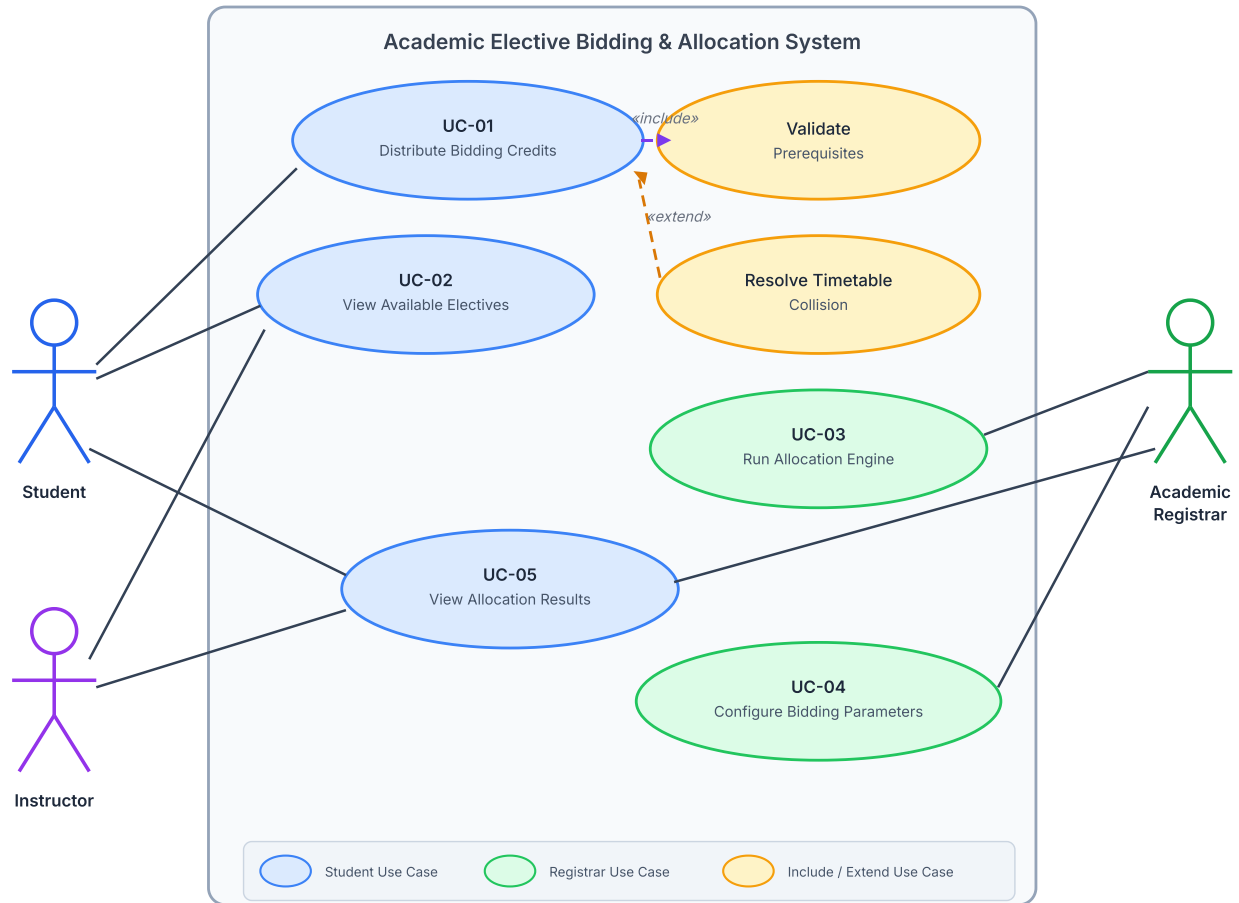


Diagram Summary

ELEMENT	NAME	ACTOR(S)	RELATIONSHIP
UC-01	Distribute Bidding Credits	Student	—
UC-02	View Available Electives	Student, Instructor	—
UC-03	Run Allocation Engine	Academic Registrar	—
UC-04	Configure Bidding Parameters	Academic Registrar	—
UC-05	View Allocation Results	Student, Academic Registrar, Instructor	—
—	Validate Prerequisites	—	«include» from UC-01
—	Resolve Timetable Collision	—	«extend» to UC-01

DELIVERABLE 3

Use-Case Flow Specification

UC-01 — Distribute Bidding Credits

Use Case ID	UC-01
Use Case Name	Distribute Bidding Credits
Primary Actor	Student
Description	A student distributes 100 bidding credits across ranked elective preferences. The system validates prerequisites and checks for timetable collisions before confirming the bid submission.

Preconditions

1. The student is authenticated and logged into the Elective Bidding Portal.
2. The Academic Registrar has opened the bidding window (current date/time falls within the configured bidding period).
3. The student has not yet finalised their bid for this bidding cycle (bid status is "Draft" or "Not Started").
4. The elective catalogue has been published with at least one available course.

Postconditions

1. The student's bidding credits are recorded against the selected elective courses in the system database.
2. Total allocated credits equal exactly 100.
3. All selected courses have validated prerequisites marked as satisfied for this student.
4. No two selected courses have overlapping timetable slots.
5. The bid status is updated to "Submitted" and a confirmation is displayed.

Main Success Scenario

STEP	ACTOR	ACTION
1	Student	Navigates to the "Elective Bidding" page from the student dashboard.
2	System	Displays the list of available elective courses with course name, instructor, schedule, available seats, and prerequisites.
3	Student	Reviews the catalogue and selects elective courses of interest (up to a maximum of 6 preferences).
4	System	For each selected course, validates that the student has completed the required prerequisite courses. (<i>«include» Validate Prerequisites</i>)
5	System	All prerequisite checks pass. The system presents a credit allocation form showing the selected courses with input fields for credit distribution.
6	Student	Distributes bidding credits across the selected courses (e.g., 40 to Course A, 30 to Course B, 20 to Course C, 10 to Course D). The total must equal exactly 100.
7	System	Validates that the total allocated credits sum to 100 and that no single course exceeds the maximum per-course credit limit set by the Registrar.
8	System	Checks the timetable schedules of all selected courses for overlapping time slots.
9	System	No timetable collisions are detected. The system displays a confirmation summary showing selected courses, credit allocations, and schedules.
10	Student	Reviews the summary and clicks "Submit Bid."
11	System	Persists the bid to the database, sets the bid status to "Submitted," and displays a confirmation message with a bid reference number.

Alternate Flows

AF-1: Prerequisite Not Met (branches from Step 4)

4a.1	System	Detects that the student has not completed one or more prerequisite courses for a selected elective.
4a.2	System	Displays a warning message: <i>"You have not completed the prerequisite [Course X] for [Elective Y]. This course cannot be included in your bid."</i>
4a.3	Student	Removes the ineligible course from the selection or replaces it with another elective.
4a.4		Flow resumes at Step 4 for the updated selection.

AF-2: Credits Do Not Sum to 100 (branches from Step 7)

7a.1	System	Detects that the total allocated credits do not equal 100 (e.g., total is 90 or 110).
7a.2	System	Displays an error message: <i>"Total credits must equal exactly 100. Current total: [X]. Please adjust your allocations."</i>
7a.3	Student	Adjusts the credit distribution.
7a.4		Flow resumes at Step 7 .

AF-3: Timetable Collision Detected (branches from Step 8)

8a.1	System	Detects that two or more selected courses have overlapping time slots (e.g., Course A and Course C both at Mon 10:00–11:00). («extend» <i>Resolve Timetable Collision</i>)
8a.2	System	Displays a collision warning: <i>"Schedule conflict detected between [Course A] and [Course C] on [Mon 10:00–11:00]. Please remove one of the conflicting courses."</i>
8a.3	Student	Removes one of the conflicting courses or replaces it with a non-conflicting alternative.
8a.4		Flow resumes at Step 4 (prerequisites re-validated for the updated selection).